DOCUMENT 1: MATH101 Syllabus

Department of Mathematics — Fall 2026

MATH101: Calculus I (4 Credits)

Course Overview & Logistics

- **Instructor:** Dr. Aris Thorne (athorne@univ.edu)
- **Lecture Schedule:** Mon/Wed/Fri 10:00 AM – 11:15 AM, Science Hall 204
- **Office Hours:** Tuesdays 2:00 PM – 4:00 PM & Thursdays 10:00 AM – 12:00 PM (Math Building, Room 312)
- **Teaching Assistant:** Elena Rostova (erostova@univ.edu) — TA Office Hours: Wed 3:00 PM – 5:00 PM
- **Prerequisites:** Satisfactory score on the Mathematics Placement Exam or MATH 099.

Course Description

An introduction to single-variable calculus. Topics include limits, continuity, derivatives, applications of differentiation (curve sketching, optimization, related rates), definite and indefinite integrals, and the Fundamental Theorem of Calculus.

Required Textbook

- *Calculus: Early Transcendentals*, 9th Edition by James Stewart.
- **Required Chapter Coverage:** Chapters 1, 2, 3, 4, 5, and 7. *(Note: Chapter 8 is excluded from the curriculum and will not be covered or tested).*

Grading Scheme

Final course grades are calculated based on the following weighted scale:

- **Quizzes (20%):** Given bi-weekly in Friday discussions. Drops the lowest score.
- **Homework Assignments (20%):** Submitted weekly via the online portal by Sunday 11:59 PM.
- **Midterm Exam (25%):** In-class exam on **October 25, 2026**.
- **Final Exam (35%):** Comprehensive final exam on **December 15, 2026 (8:00 AM – 11:00 AM)**.

Grading Scale: A (93-100%), A- (90-92%), B+ (87-89%), B (83-86%), B- (80-82%), C+ (77-79%), C (73-76%), C- (70-72%), D (60-69%), F (<60%).

Weekly Schedule & Topics

- **Week 1 (Aug 31 – Sep 4):** Review of Functions, Tangent Lines, and Rate of Change (Stewart Ch 1.1–1.3)
- **Week 2 (Sep 7 – Sep 11):** The Limit of a Function, One-Sided Limits, and Limit Laws (Stewart Ch 2.1–2.3)
- **Week 3 (Sep 14 – Sep 18):** Continuity, Infinite Limits, and Asymptotes (Stewart Ch 2.5–2.6)
- **Week 4 (Sep 21 – Sep 25):** Derivatives, Velocity, and Rates of Change (Stewart Ch 2.7–2.8)
- **Week 5 (Sep 28 – Oct 2):** Differentiation Rules: Power, Product, and Quotient Rules (Stewart Ch 3.1–3.3)
- **Week 6 (Oct 5 – Oct 9):** Derivatives of Trigonometric Functions and the Chain Rule (Stewart Ch 3.4–3.5)
- **Week 7 (Oct 12 – Oct 16):** Implicit Differentiation and Logarithmic Differentiation (Stewart Ch 3.6–3.8)
- **Week 8 (Oct 19 – Oct 23):** Related Rates and Applications of Differentiation (Stewart Ch 3.9, 4.1)
    - *Midterm Exam: October 25, 2026*
- **Week 9 (Oct 26 – Oct 30):** Maximum and Minimum Values, Mean Value Theorem, Curve Sketching (Stewart Ch 4.2–4.5)
- **Week 10 (Nov 2 – Nov 6):** Optimization Problems and Newton's Method (Stewart Ch 4.7–4.8)
- **Week 11 (Nov 9 – Nov 13):** Antiderivatives, Definite Integrals, and the Fundamental Theorem of Calculus (Stewart Ch 4.9, 5.1–5.3)
- **Week 12 (Nov 16 – Nov 20):** The Substitution Rule and Integration by Parts (Stewart Ch 5.5, 7.1)
- **Week 13 (Nov 23 – Dec 4):** Applications of Integration: Area Between Curves (Stewart Ch 6.1) & Final Review

Course Policies

- **Late Submission Policy:** Homework assignments submitted late will incur a **15% penalty per day** up to a maximum of 3 days. Submissions more than 72 hours past the deadline will receive a score of zero. No extensions are granted without a verified medical or university excuse submitted at least 24 hours in advance.
- **Make-up Exams:** Make-up exams for the Midterm or Final are granted ONLY for officially documented emergencies. You must notify Dr. Thorne prior to the exam start time.
- **Academic Integrity:** Collaboration on homework is encouraged for brainstorming, but all written work must be produced independently. Calculators are permitted on homework but restricted during quizzes and exams (only basic non-graphing calculators allowed).